

=====

ACME PETROCHEMICAL DEMO COMPLEX - UNIT 2300

PROCESS FLOW DIAGRAM (PFD): DECOMPOSER REACTOR & NEUTRALIZATION SECTION

Drawing Number: PFD-23-0005 | Revision: Z1 | Process Unit: Unit 2300

1. PROCESS TOPOLOGY & REACTION LOOP

- Concentrated 82 wt% CHP feed (Stream S229 from V-2301 / P-2301A/B) and dilute sulfuric acid catalyst (Stream S378 from V-2302) enter the reactor (R-2301) through the top feed.
- High-rate circulation loop (Stream S379, 185,000 kg/hr, 12.7:1 recycle ratio) circulates reaction mixture from the reactor (R-2301) through the bottom effluent line to the separator (D-2304) and back to the reactor (R-2301) through the recycle line.
- Cleaved Phenol + Acetone reaction product (Stream S384, 14,500 kg/hr) exits D-2304 via Level Control Valve LV-1304 to the storage tank (T-2305).
- Emergency Quench Header (Stream S387) supplies chilled Cumene through SIL-2 Valve XV-1309 directly into D-2304.

=====									
PFD-23-0005 PAGE 2 OF 2 - HEAT & MATERIAL BALANCE TABLE (DECOMPOSER SECTION)									
Stream ID	From -> To	Fluid Service	Mass Flow (kg/hr)	Temp (degC)	Pressure (kg/cm2g)	Key Composition			
S229	V-2301 / P-2301 -> D-2304	Concentrated CHP Feed	14,500	92.0	6.8	82.0 wt% CHP, 13.0 wt% Cumene			
S310	Catalyst Pkg -> D-2304	Dilute H2SO4 Catalyst	45	35.0	4.5	180 ppmw H2SO4 in Acetone			
S379	D-2304 -> E-2307	Reactor Recycle Draw	185,000	82.0	4.2	49.5 wt% Phenol, 30.8 wt% Acetone, <0.8			
S380	E-2307 -> D-2304	Cooled Recycle Return	185,000	74.0	3.1	49.5 wt% Phenol, 30.8 wt% Acetone			
S384	D-2304 -> D-2310	Cleaved Crude Product	14,545	82.0	1.8	49.5 wt% Phenol, 30.8 wt% Acetone, 2.2			
S387	Quench Drum -> D-2304	Emergency Cumene Quench	0 (Standby) / 25,000 (Trip)	30.0	8.5	99.5 wt% Pu			